

Chapter 18

Lecture 18: One-point compactification. Criteria for local compactness.

Theorem 18.0.1. *Let X be a topological space. Then X is Hausdorff locally compact if and only if there exists a space Y such that*

1. X is a subspace of Y .
2. $Y \setminus X$ contains exactly one point p .
3. Y is compact Hausdorff.

This Y is unique in the following sense: If Y, Y' are two spaces with the above properties and $Y = X \cup \{p\}, Y' = X \cup \{q\}$, then there exists a homeomorphism $h : Y \rightarrow Y'$ such that $h(x) = x$ for all $x \in X$ and $h(p) = q$.

Proof

Uniqueness Let $Y = X \cup \{p\}, Y' = X \cup \{q\}$ satisfies the above properties. Define $h : Y \rightarrow Y'$ as follows: $h(x) = x$ for all $x \in X$ and $h(p) = q$. We show that h is continuous. But the function is symmetric, so it is enough to show that $h(U)$ is open in Y' for all open set U in Y .

Take U be open. There are two cases.

If $U \subseteq X$, then we are done.

Suppose $p \in U$. Then $C = Y \setminus U$ is closed in Y . So C is compact. So $h(C) = C$ is compact in Y' . Since Y' is Hausdorff, C is closed in Y' . So $Y' \setminus C$ is open in Y' . Note that $h(U) = Y' \setminus C$. So $h(U)$ is open in Y' .

Construction We introduce the topology on $Y = X \cup \{\infty\}$ as follows:

There are two types of open sets in Y :

1. $U \subseteq X \subseteq Y$ is open in X .
2. $U = Y \setminus C$ where C is compact subspace of X .

We need to check that this is a topology.

For the intersection of two open sets of type 1, we have $U_1 \cap U_2$ is open in X thus is open in Y .

For the intersection of type 1 and type 2, we have $U_1 \cap (Y \setminus C) = U_1 \cap (X \setminus C)$ which is the union of open sets in X thus is open in Y because C is closed in X (as a compact subspace of a Hausdorff space).

For the intersection of two open sets of type 2, we have $(Y \setminus C_1) \cap (Y \setminus C_2) = Y \setminus (C_1 \cup C_2)$ which is open in Y because $C_1 \cup C_2$ is compact.

Similarly, one checks that the union of any collection of open sets is open. We have

1. $\bigcup U_\alpha = U$ is of type 1.
2. $\bigcup (Y \setminus C_\alpha) = Y \setminus \bigcap C_\alpha = Y \setminus C$ where $C = \bigcap C_\alpha$ is compact (Easy to check). This is of type 2.
3. $\bigcup U_\alpha \cup \bigcup (Y \setminus C_\beta) = U \cup (Y \setminus C) = Y \setminus (C \setminus U)$. This is of type 2 because $C \setminus U = C \cap (X \setminus U)$ is closed in C thus is compact because a closed subset of a compact set is compact.

Now we show that X is a subspace of Y . Given any open set V of Y , we show its intersection with X is open in X . If V is of type 1, then $V \cap X = V$ which is open in X . If V is of type 2, then $V \cap X = (Y \setminus C) \cap X = X \setminus C$ which is open in X because C is closed in X as a compact subspace of a Hausdorff space. Conversely, given any open set U of X , U is an open set of Y of type 1. So X is a subspace of Y .

We show that Y is compact.

Let $\mathcal{A}$ be an open covering of Y . The collection $\mathcal{A}$ must contain an open set containing ∞ . So there exists a compact set C such that $U = Y \setminus C$ with $U \in \mathcal{A}$. Take all the members of $\mathcal{A}$ different from U and intersect them with X . This gives a collection of open sets in X which covers C . Since C is compact, there exists a finite subcovering of C , say $\{V_1, V_2, \dots, V_n\}$. Then $\{U, V_1, V_2, \dots, V_n\}$ is a finite subcovering of Y . So Y is compact.

We show that Y is Hausdorff.

Take two distinct points $x, y \in Y$. There are two cases.

1. If $x, y \in X$, since X is Hausdorff, there exist disjoint open sets U, V in X such that $x \in U, y \in V$. Then U, V are open in Y and disjoint.
2. If one of them is ∞ , say $y = \infty$. Since X is locally compact at x , there exists an open neighborhood U of x and a compact set C such that $U \subseteq C$. So C is closed in X thus in Y . Then $V = Y \setminus C$ is an open neighborhood of ∞ . Clearly, $U \cap V = \emptyset$.

Finally we prove the other direction. Suppose a space Y satisfying conditions (1)-(3) exists. Then X is Hausdorff as a subspace of a Hausdorff space. Given any $x \in X$, we show X is locally compact at x . Choose disjoint open sets U, V in Y such that $x \in U$ and $p \in V$. Then $C = Y \setminus V$ is compact as a closed subset of a compact space. Also, $x \in U \subseteq C$. So X is locally compact at x . ■

Definition 18.0.1. If Y and X are as in the above theorem, then Y is called the one-point compactification of X .

Example 18.0.1. 1. The one point compactification of $\mathbb{R}$ is homeomorphic to S^1 . We denote $\overline{\mathbb{R}} \cong S^1$.

2. The one point compactification of $\mathbb{R}^n$ is homeomorphic to S^n . We denote $\overline{\mathbb{R}^n} \cong S^n$.

3. The one point compactification of $\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ (Riemann sphere) is homeomorphic to S^2 .

Theorem 18.0.2. Let X be a Hausdorff space. Then X is locally compact if and only if for any $x \in X$ and any neighborhood U of x , there exists a neighborhood V of x such that $\bar{V}$ is compact and $\bar{V} \subseteq U$.

Proof

$\Leftarrow$ $x \in V \subseteq \bar{V} \subseteq U$ and $\bar{V}$ is compact. So X is locally compact at x .

$\Rightarrow$ Let Y be the one-point compactification(compact and Hausdorff) of X . Let U be a neighborhood of x in Y . Then $Y \setminus U := C$ is a closed subset of Y . So C is compact subspace of Y (Hausdorff). By lemma 13.0.2, there exists two open sets V, W in Y such that $x \in V, C \subseteq W$ and $V \cap W = \emptyset$. Then $\bar{V}$ is compact and $\bar{V} \cap C = \emptyset$. So $\bar{V} \subseteq U$. ■

Corollary 18.0.1. Let X be locally compact Hausdorff space. $A \subseteq X$ is open or closed. Then A is locally compact.

Proof

Let $A \subseteq X$ be closed. Given $x \in A$, since X is locally compact, there exists an open neighborhood U of x in X and a compact set C such that $x \in U \subseteq C$. Then $C \cap A$ is closed in C thus is compact, and it contains the neighborhood $U \cap A$ of x in A . So A is locally compact at x .

Let $A \subseteq X$ be open. Given $x \in A$, since X is locally compact and X is Hausdorff, by the previous theorem, there exists a neighborhood V of x such that $\bar{V}$ is compact and $\bar{V} \subseteq A$. Then $C = \bar{V}$ is a compact set containing the neighborhood V of x in A . So A is locally compact at x . ■

Corollary 18.0.2. X is locally compact Hausdorff if and only if X is homeomorphic to an open subspace of a compact Hausdorff space.

Exercise Show the above corollary by 18.0.1 and 18.0.1.

Solution

$\Rightarrow$ Let Y be the one-point compactification of X . Recall that Hausdorff is T_1 . So $X = Y \setminus \{p\}$ is open in Y . The identity map $id: X \rightarrow X$ is a homeomorphism from X to the open subspace X of Y .

$\Leftarrow$ Let X be homeomorphic to an open subspace U of a compact Hausdorff space Z . Since Z is compact Hausdorff, Z is locally compact Hausdorff. By 18.0.1, U is locally compact Hausdorff. Since X is homeomorphic to U , X is locally compact Hausdorff. ■

18.1 Urysohn's Metrization Theorem

Theorem 18.1.1 (Our goal in the next few lectures). *Every X that is regular(T_3) and second-countable is metrizable.*